

ECON 4370

Building AI Economic Data Agents

Dr. Fidel González

Where We Are in the Course

You already know how to:

- Pull data from APIs (FRED, BLS)
- Clean and merge datasets
- Compute transformations (YoY, growth rates)
- Create visualizations
- Compute correlations

Today we add one layer:

Automation + Decision Logic = AI Agent

Traditional Script vs AI Agent

Traditional Python Script

- Static
- Hard-coded variables
- User must know Python
- Limited flexibility

AI Agent

- Interprets natural language
- Chooses tools dynamically
- Executes multi-step workflow
- Generates interpretation

What Is an AI Agent?

User Question



LLM Planner (JSON plan)



Python Tools (Data + Analysis)



Plot + Statistics + Narrative

Characteristics:

- Structured planner
- Limited decision space
- Reliable and controlled
- Good for learning architecture

Goal: Understand how LLM + tools interact.

Version 2: Autonomous (Intermediate) Agent

Upgrades:

- Maps concepts → FRED series
- Chooses transformations (YoY vs level)
- Selects tasks (plot, correlation)
- Adapts to new questions

Still safe because:

- Approved series dictionary
- Structured JSON schema
- Controlled tool set

What Is an API Key?

An API key is:

- A secret string tied to your account
- Used for authentication
- Used for billing

Example format:

`sk-xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx`

It identifies your account for every API request.

Important: ChatGPT Plus vs API

ChatGPT Plus

- Subscription for chat interface

OpenAI API

- Separate billing
- Usage-based pricing
- Required for programmatic access

Security Best Practices

- Never hard-code API keys in shared notebooks
- Never upload keys to GitHub
- Use environment variables
- Revoke keys if exposed

Data Engineering Lesson: Frequency Mismatch

Example:

- CPI → Monthly
- Oil prices → Daily

If merged directly:

- Mostly missing values
- Correlation = NaN

Solution:

- Resample oil to monthly average
- Then merge

Interactive Visualization: Plotly

Why switch to Plotly?

- Zoomable
- Hover values
- Professional dashboard feel
- Better for demos

Interactive graphs $\Rightarrow$ More engaging analysis.

Live Demo Script (In Class)

Try these:

- “Is there evidence of a Phillips Curve since 1990?”
- “Compare oil prices and inflation after 2015.”
- “Did wages keep up with inflation after COVID?”
- “Compare federal funds rate and inflation since 2000.”

Observe:

- Series selection
- Transformations
- Plot generation
- Correlation output

Why This Matters for Business

Agents allow:

- Executives to ask natural language questions
- Automated economic dashboards
- AI-assisted decision systems
- Workflow automation

This is how AI integrates into organizations.

Final Takeaway

A script follows instructions.

An agent decides which instructions to follow.

Baseline = Structured Automation

Autonomous = Adaptive Intelligence

Both rely on solid data skills.

Next Steps

Next enhancements could include:

- Automatic regression tool
- Forecasting module
- Dashboard interface (Streamlit)
- Multi-step economic reasoning

You are now building real AI systems.